Explain the difference between Frontend, Backend, and Full-Stack Development with suitable examples.

Answer

Frontend Development

- Frontend is the part of a website that users can see and use.
- It is made using **HTML, CSS, and JavaScript**.

Backend Development

- Backend works behind the scenes.
- It manages data, users, and the server.

Full-Stack Development

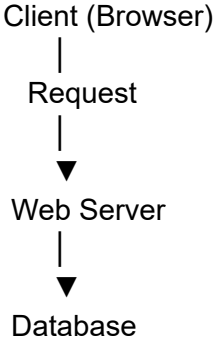
- A full-stack developer works on both frontend and backend.

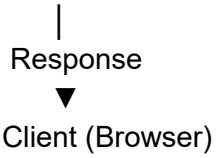
Example (Amazon Website):

- **Frontend:** Product images, search bar, Add to Cart button.
- **Backend:** Stores product details, user accounts, and orders.
- **Full-Stack:** Builds both frontend and backend.

Q2. Create a simple diagram showing the Client–Server Model.

Answer





Explanation

- The client sends a request.
- The server receives the request.
- The server gets data from the database.
- The server sends the response back to the client.

Q3. Describe how a browser requests and displays a web page.

Answer

1. The user enters a website address (URL).
2. The browser sends a request to the server.
3. The server receives the request.
4. The server finds the required webpage.
5. The server sends the webpage to the browser.
6. The browser displays the webpage on the screen.

Example: Opening **www.youtube.com**.

Q4. List the tools required for a web development environment and explain their purpose.

Answer

Tool	Purpose
VS Code	Write and edit code
Chrome	Test websites

HTML	Create webpage structure
CSS	Design the webpage
JavaScript	Add interactivity
Node.js	Run JavaScript
Git	Manage code versions
GitHub	Store code online
Live Server	Auto-refresh the webpage

Q5. Explain what a Web Server is. Give examples.

Answer

A **Web Server** is software or a computer that stores websites and sends web pages to users when they request them.

Examples

- Apache
- Nginx
- Microsoft IIS
- LiteSpeed
- Node.js Server

Q6. Define the roles of Frontend Developer, Backend Developer, and Database Administrator.

Answer

Frontend Developer

- Creates webpage design.
- Makes buttons, menus, and forms.

Backend Developer

- Creates server-side programs.
- Connects the website with the database.

Database Administrator (DBA)

- Stores and manages data.
- Keeps the database secure.

Q7. Explain how to install and configure VS Code for HTML, CSS, and JavaScript.

Answer

1. Download and install **VS Code**.
2. Open VS Code.
3. Install extensions:
    - Live Server
    - HTML CSS Support
    - Prettier
4. Create files:
    - `index.html`
    - `style.css`
    - `script.js`
5. Open `index.html` using **Live Server**.

Q8. Explain the difference between Static and Dynamic Websites.

Answer

Static Website	Dynamic Website

Content does not change.	Content changes according to the user.
No database is required.	Uses a database.
Faster and simpler.	More interactive.

Examples

- **Static:** Portfolio website
- **Dynamic:** Facebook, Amazon

Q9. List five web browsers and explain their rendering engines.

Answer

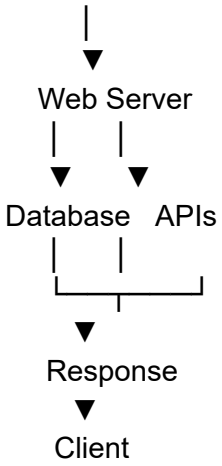
Browser	Rendering Engine
Google Chrome	Blink
Microsoft Edge	Blink
Mozilla Firefox	Gecko
Apple Safari	WebKit
Opera	Blink

Rendering Engine: It reads HTML, CSS, and JavaScript and displays the webpage.

Q10. Draw the basic Web Architecture Flow.

Answer

Client
|
Request



Explanation

- The **Client** sends a request.
- The **Server** processes it.
- The **Database** stores data.
- **APIs** help different applications communicate.
- The server sends the response back to the client.